

Content

Learning objectives and content

Objectives

This course introduces the student to nanoscale electronic and photonic semiconductor devices. We will present the relevant semiconductor and device physics and discuss the miniaturization and integration of components that form the basis of today's information society.

The main learning outcomes in this course are that the student will:

- understand the basic mechanisms and description of charge transport in intrinsic and doped semiconductors;
- be able to apply those mechanisms to describe p-n junctions, and metal semiconductor transitions;
- understand the operation of CMOS field effect transistors that form the basis of integrated circuits;
- understand the operation of semiconductor photonic devices solar cells, detectors, light emitting diode and laser;
- understanding the issue of reducing the size of semiconductor based devices to the nanometer scale and the role that nanoscale structures play.

Content

The following subjects will be discussed: Semiconductor basics, intrinsic and extrinsic semiconductors, carrier transport, carrier recombination and generation, p–n Junction, MOS Capacitors MOS Field Effect Transistor

Nanoelectronics: CMOS integration technology

Nanophotonics: confined semiconductor devices, nanolasers.

2025

Nano devices and integration (5XPB0)

Practical information

Course module

5XPB0

Course type

Bachelor College

Credits (ECTS)

5 EC

Category

Advanced

Language of instruction

English

Offered by

Electrical Engineering - Photonic Integration Group

Remark

Last year subject has been taught: 25/26

Last examination year: 25/26

Replaced by course 5XPF0 as of 2026/2027

General course information

This is the third course in the Nano-Science and technology track. Due to overlap of this course with 5CCA0 "Semiconductor physics and materials", Master students who have already passed the exam 5CCA0 are strongly advised not to choose the course 5XPB0 as an elective.

Short promotional description of the course

This course will introduce the student to nanoscale electronic and photonic devices, and the integration of such devices to form useful circuits. We will discuss how miniaturization affects the circuits and how circuit requirements drive the miniaturization of its components.

Course enrolment

OSIRIS Student

You can register yourself via OSIRIS Student. To do so, go to Enroll / Course.

Enrolment periods

- Block 3

Timeslot(s)

C - Tu 1-4, Fr 5-8

Enrolment period

15 November 2025 until 4 January 2026 23:59

Unenrolment

until 2 February 2026 23:59

Entrance requirements

Assumed previous knowledge

2WBB0 Calculus variant B

3ENX0 Nanomaterials: P&C

3NBB0 Applied physical sciences formal

Instructional modes

Instructional modes

- Lecture

Remark

- Lecture

Remark

Tests

Tests

- Assignment

Interim examination

Test weight

30

Minimum grade

- Written examination

Final examination

Test weight

70

Minimum grade

5.0

Test enrolment

OSIRIS Student

You can register yourself via OSIRIS Student. To do so, go to Enroll / Test.

Assignment

- Block 3

Enrolment period

15 November 2025 until 22 March 2026 23:59

Unenrolment

until 22 March 2026

Written examination

- Block 3

Enrolment period

15 November 2025 until 22 March 2026 23:59

Unenrolment

until 22 March 2026

Test date

15 April 2026

- Block 4

Enrolment period

15 November 2025 until 7 June 2026 23:59

Unenrolment

until 7 June 2026

Test date

26 June 2026

Lecturer(s)

Co-lecturer

- prof.dr.ir. E. Cantatore

Contactperson for the course

- prof.dr.ir. M.J.R. Heck

Responsible lecturer

- prof.dr.ir. M.J.R. Heck

Subject matter expert

- prof.dr.ir. M.J.R. Heck

Materials

Required material

- Semiconductor Physics and Devices, Fourth Edition" Donald Neamen, McGraw Hill, ISBN: 978-007-108902-9 (3rd edition can also be used)
- Slides on CANVAS

Do you want to know more?

prof.dr.ir. M.J.R. Heck can tell you more about this course.

Send an e-mail to m.heck@tue.nl